



Disentangled Anomaly Detection For Multivariate Time Series

Xin Jie
jiexin@zju.edu.cn
Zhejiang University

Xixi Zhou
xixi.zxx@zju.edu.cn
Zhejiang University

Chanfei Su
suchanfei@oppo.com
OPPO Research Institute

Zijun Zhou
zhouzijun1@oppo.com
OPPO Research Institute

Yuqing Yuan
yuanyuqing@oppo.com
OPPO Research Institute

Jiajun Bu
bjj@zju.edu.cn
Zhejiang University

Haishuai Wang
haishuai.wang@zju.edu.cn
Zhejiang University

ABSTRACT

Anomaly detection in time series that aims to identify unusual patterns has attracted a lot of attention recently. However, the representation of abnormal and normal data is difficult to be distinguished because they are usually entangled. Recently, disentanglement theory based on variational auto-encoder (VAE) has shown great potential in machine learning and achieved great success in computer vision and natural language processing. In this paper, we propose a novel disentangled anomaly detection approach that adopts VAE-based disentanglement networks for anomaly detection in multivariate time series. The proposed method learns high-quality disentangled latent factors in a continuous representation space to facilitate the identification of anomalies from normal data. Extensive experiments demonstrate that our proposed lightweight model DA-VAE achieves state-of-the-art performance.

CCS CONCEPTS

• Security and privacy → Intrusion/anomaly detection and malware mitigation; • Computing methodologies → Learning latent representations.

KEYWORDS

anomaly detection, time series, VAE, disentanglement

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1 INTRODUCTION

In recent years, numerous time series data analysis methods have been proposed and successfully applied to different application fields, such as risk prediction and anomaly detection tasks. Among

them, anomaly detection is a core task that has been widely studied for time series data, and anomaly detection on time series data becomes more and more important in many application scenarios such as cloud service anomaly detection, industry control system monitoring, and the status anomaly detection of terminal devices of the Internet of things.

Traditional approaches in anomaly detection are based on density estimation and clustering technologies. These traditional approaches neglect temporal information and face challenges in their applicability to novel real-world scenarios. On the other hand, with the development of deep learning models, semi-supervised/unsupervised methods have been widely adopted, which learn normal data structures and effectively solve the problem of insufficient labeling in anomaly detection tasks.

The RNN-based and LSTM-based models are very popular in recent years. This type of model can model time series data with time dependence, and its variants, LSTM and GRU, can learn long-term dependencies in the data. The LSTM-VAE model combines the LSTM architecture for temporal modeling with the Variational AutoEncoder (VAE) [7] for reconstruction. OmniAnomaly [13] takes the GRU model [3] with a normalizing flow, utilizing reconstruction probabilities for anomaly detection. InterFusion, a hierarchical VAE model, captures inter- and intra-dependencies among multiple series simultaneously. After the transformer is presented, several approaches [14, 19] are proposed for time series analysis by the attention architecture. The graph structure is widely used to represent complex inter-sensor relationships in time series analysis tasks, including anomaly detection [2, 22]. These methods have complex network structures and high computational costs, but their performance benefits are often insignificant. Additionally, other approaches leveraging traditional feature-extracting methods and more simple neural network structures achieve great success in time series tasks [15, 16].

The current mainstream deep learning methods for time series anomaly detection are mainly reconstruction-based models [17, 25]. This model type performs unsupervised learning on normal data with the goal of reconstruction. During detection, difficult to reconstruct samples will be identified as anomalies. This method is simple and can quickly identify anomalies when applied, but it is difficult to distinguish the representation of anomalies and normal data, so its performance is limited. Another class of methods is the correlation-difference-based method [19]. This model type must build a standard to distinguish anomalies from normal data, usually to characterize the difference in representation. Although the effect is better, it is relatively complex, and the identification

* Xin Jie and Xixi Zhou contributed equally to this work.

† Corresponding authors: Haishuai Wang and Jiajun Bu.

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of anomalies is slow because they need to pay attention to the contextual information of more detection points.

To address the above issues, in this paper, we introduce the VAE-based disentanglement representation learning (DRL) into time series data to learn a more interpretable anomaly detection model. From a perspective disentanglement representation learning (DRL), VAE aims to separate different interpretable factors in the observed data and encode them into independent and distinct latent variables in the representation space, where all the disentangled latent representations are aligned with real semantics [7]. We adopt a VAE-based disentanglement framework as the backbone network and control the disentanglement capability of the model through parameter β [4]. Our proposed model brings two benefits. 1) The model is based on unsupervised training and can reconstruct data well, so it can quickly identify anomalies. 2) Disentanglement representation naturally widens the representation distance between abnormal and normal data without introducing additional training strategies, which enhances the anomaly detection performance of time series data. In short, disentangled networks can produce better latent representations—the continuous space is conducive to representationally distinguishing different input data modalities. We use a simple linear network as the decoder of VAE to avoid the KL vanishing problem when applying powerful autoregressive decoders to time series data. Our method is simple to train while outperforming other strong reconstruction-based baselines.

The main contributions of this paper are summarized as follows:

- (1) We present a lightweight model that integrates the VAE-based disentanglement for anomaly detection.
- (2) We regularize the representation space and reconstruction quality in the VAE architecture by controlling the disentanglement capability of the model. Our model shows that the disentanglement space is elegantly and effectively suitable for the anomaly detection task.
- (3) A large number of experiments show that our proposed method significantly overpasses over the strong baseline models of time series anomaly detection, and our proposed DA-VAE achieves state-of-the-art performance for four wide-studied time series datasets.

2 METHODOLOGY

This section introduces our proposed VAE-based disentangled model for multivariate time series anomaly detection. We first provide the problem statement on anomaly detection (Section 2.1). Meanwhile, we adopt the VAE-based disentanglement learning framework to regularize the normalized representation space, enhancing anomaly detection performance on time series data (Section 2.2). Finally, we introduce the anomaly scoring function we use (Section 2.3).

2.1 Problem Formulation

We focus on anomaly detection on multivariate time series $X_{1:T}^M$, defined as $x_{1:T}^i = \{x_1^i, x_2^i, \dots, x_T^i\}$, where $i \in \{1, 2, \dots, M\}$. M denotes the dimension of the multivariate time series, and T is the maximum time step. The input data in training is divided into N windows, denoted as $w = \{X_j^M, \dots, X_{j+(K-1)}^M\}$, where K is the length of each time window. We aim to output a vector $y \in \mathbb{R}^K$ that denotes the anomaly detection results of the K points in a window.

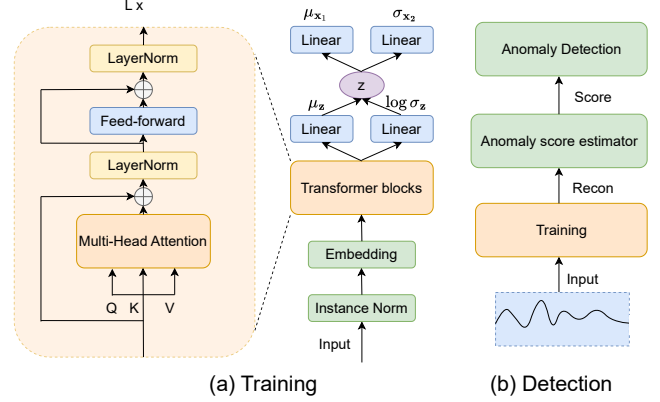


Figure 1: The architecture of our proposed VAE-based disentangled anomaly detection (DA-VAE) model.

2.2 Overall Architecture

Figure 1 shows our VAE-based disentangled model, which contains two parts: encoder and decoder. The encoder adopts the transformer architecture widely used in time series modeling. To prevent the KL collapse problem when applying the VAE architecture in time series tasks [8], we do not operate a powerful autoregressive decoder but choose the most basic linear network. In recent years, much literature has confirmed through extensive experiments that linear predictors can achieve better results [10, 20].

(1) Normalization and Input Embedding In our approach, we adopt the normalization-denormalization presented in RevIN [6] and No-stationary transformers [11], which propose a similar approach using a symmetric structure to remove statistical features from the input of the time series and restore them from the output. This strategy can effectively improve the modeling ability of the time series model by smoothing the statistical characteristics (e.g., the mean and variance) of time series data that change over time. Then, the output time series ($X \in \mathbb{R}^{M \times K}$) will go through an embedding layer and serve as the encoder's input ($X \in \mathbb{R}^{M \times d_{model}}$).

(2) Transformer Encoder We use powerful transformer as the encoder. At the same time, to allow the model to have multivariate modeling capabilities without introducing additional graph structures, we draw on the success of iTransformer [10] and use the attention mechanism to capture multivariate correlations. After obtaining the embedding $X \in \mathbb{R}^{M \times d_{model}}$, the calculation process of the self-attention in l -th layer ($l = 0, 1, \dots, L - 1$) is described as follows:

$$Q, K, V = X^{l-1} W_Q^l, X^{l-1} W_K^l, X^{l-1} W_V^l \quad (1)$$

$$\mathcal{H}^l = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{model}}}\right)V \quad (2)$$

$$X^l = \text{Feed-forward}(\mathcal{H}^l), \quad (3)$$

where $Q, K, V \in \mathbb{R}^{M \times d_{model}}$, represent the query, key, value of self-attention. $W_Q^l, W_K^l, W_V^l \in \mathbb{R}^{d_{model} \times d_{model}}$ represent the parameter matrices for Q, K, V . Feed-forward represents a learnable fully connected layer. Finally, we can obtain the representation extracted

Table 1: The experimental results of Precision, Recall, and F1-Score(%) on baselines and our model (DA-VAE). We highlight the best and second-best results in bold and underlined styles.

	MSL			SMAP			PSM			SWaT			(Avg.)
	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	Precision	Recall	F1	F1
Transformer	71.57	87.37	78.68	89.37	57.12	69.70	62.75	<u>96.56</u>	76.07	68.84	96.53	80.37	76.21
Informer	81.77	86.48	84.06	90.11	57.13	69.92	64.27	96.33	77.10	70.29	<u>96.75</u>	81.43	78.13
Autoformer	77.27	80.92	79.05	90.40	58.62	71.12	99.08	88.15	93.29	89.85	95.81	92.74	84.05
FEDformer	77.14	80.07	78.57	90.47	58.10	70.76	97.31	97.16	97.23	90.17	96.42	93.19	84.94
Pyraformer	83.81	85.93	84.86	<u>92.54</u>	57.71	71.09	71.67	96.02	82.08	87.92	96.00	91.78	82.45
Stationary	68.55	<u>89.14</u>	77.50	89.37	59.02	71.09	97.82	96.76	97.29	68.03	96.75	79.88	81.44
LightTS	82.40	75.78	78.95	92.58	55.27	69.21	98.37	95.97	97.15	<u>91.98</u>	94.72	<u>93.33</u>	84.66
DLinear	84.34	85.42	84.88	92.32	55.41	69.26	98.28	89.26	93.55	80.91	95.30	87.52	83.80
AT	79.61	87.37	83.31	91.85	58.11	71.18	68.35	94.72	79.40	72.51	97.32	83.10	79.25
TimesNet	<u>89.54</u>	75.36	81.84	90.14	56.40	69.39	98.51	96.20	<u>97.34</u>	90.75	95.40	93.02	85.40
FPT	82.00	82.91	82.45	90.60	60.95	72.88	98.62	95.68	97.13	92.20	96.34	94.23	86.67
DA-VAE	90.12	91.96	91.03	85.08	78.29	81.55	98.20	99.62	98.91	90.85	89.32	90.08	90.39

by the encoder:

$$\mu_z = \text{Feed-forward}(\mathcal{H}^L) \quad \text{and} \quad \sigma_z = \text{Feed-forward}(\mathcal{H}^L), \quad (4)$$

where $\mu_z, \sigma_z \in \mathbb{R}^{M \times d_z}$.

(3) **VAE Architecture** VAE is a generative model that has the inference and generative phases [7]. In its inference phase, VAE maps the data x to be latent variables z by an encoder network, which usually is modeled as a prior Gaussian distribution $N(0, I)$. After that, the reconstruction data is sampled from $p_\theta(x|z)$ in a generative phase, where a decoder network parameterizes θ . Because the posterior $p_\theta(x|z)$ is intractable, the variational inference techniques are used to approximate it with the approximation posterior $q_\phi(x|z)$, which is usually set as a distribution of $N(\mu_\phi(x), \sigma_\phi^2(x))$, where ϕ is parameterized as another neural network. The learning objective of VAE is named evidence lower bound (ELBO) in the VAE's original literature, we adopt the β -VAE [4] model and add a penalty weight β to the loss function. **A heavier β obtains better disentangled representation.** The loss function is written as below:

$$\mathcal{L} = -\mathbb{E}_q[\log p(x|z)] + \beta D_{KL}(q(z|x)||p(z)), \quad (5)$$

where the first item is the reconstruction objective and the second is the KL objective. Among them, we set the likelihood $p(x|z)$ to a Gaussian distribution.

2.3 Anomaly Detection

In evaluation, we use the negative log reconstruction probability density as the anomaly detection score, which is calculated by:

$$\text{Score} = -\log p(x|z). \quad (6)$$

Because we are doing anomaly detection and focus on distinguishing anomalies and normal data rather than the error of reconstruction, probability estimation introduces randomness and is more suitable for this task.

3 EXPERIMENTS

We evaluate our proposed DA-VAE with four widely-studied datasets. We first present the datasets, the baseline models, the evaluation metrics, and the parameter settings in our experiments. Then we

illustrate and analyze our experimental results. Our model shows good performance on multiple metrics of anomaly detection.

3.1 Datasets

(1) **Mars Science Laboratory¹ (MSL)** [5]. The dataset is collected by NASA with 55 dimensions from the sensors in the Mars rover.

(2) **Soil Moisture Active Passive² (SMAP)** [5]. The dataset is also collected by NASA with 25 dimensions.

(3) **Pooled Server Metrics³ (PSM)** [1]. The dataset is collected internally from multiple application server nodes at eBay with 26 dimensions.

(4) **Secure Water Treatment⁴ (SWaT)** [12]. The dataset used in this study comprises data from 51 sensors installed in a critical infrastructure system that operates around the clock. We use SWaT's Normal V1 version as the training set and its Attack V0 version as the test set.

3.2 Preparations

We refer to the original settings in the TimesNet [17]. For time series data, we use the sliding window to intercept a segment of subsequence. If the anomaly score at a time point is greater than a certain threshold δ , it will be marked as an anomaly. We also use the widely-used point adjustment strategy to adjust the detection results. We divide the training data into a training set and a validation set in a ratio of 0.8:0.2, and the validation set does not contain labeling information of anomaly.

3.3 Baseline Methods

We include 11 strong baselines in experiments: Transformer [14], Informer [23], Autoformer [18], FEDformer [24], Pyraformer [9], Stationary-Transformer(Stationary) [11], LightTS [21], DLinear [20], Anomaly transformer(AT) [19], TimesNet [17] and FPT [25]. For the results of these baselines, we referred to the results in the original papers of the TimesNet and the FPT. We use reconstruction error in Anomaly Transformer for fair comparison.

¹<https://github.com/thuml/Time-Series-Library>

²<https://github.com/thuml/Time-Series-Library>

³<https://github.com/thuml/Time-Series-Library>

⁴https://itrust.sutd.edu.sg/itrust-labs_datasets

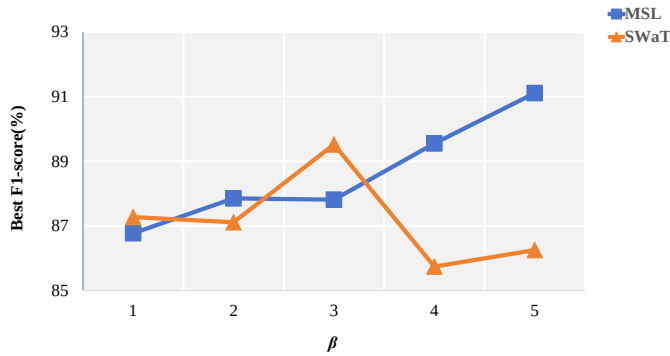


Figure 2: Hyperparameter analysis of β on MSL and SWaT.

3.4 Results Analysis

Table 1 shows the experimental results of our model, indicating the proposed model significantly improves the average performance on the four public datasets. After learning the disentangled representation, the model can greatly improve anomaly detection performance on MSL, SMAP and PSM.

We also analyze the impact of hyperparameter β on the detection results using two example datasets, as shown in the Figure 2. As β gradually increases, the model pays more attention to disentangled representations rather than reconstruction. High-quality disentangled representations can achieve better anomaly detection performance on MSL. However, on the SWaT dataset, the optimal result occurs when β is approximately equal to 3. Because the data in SWaT is more complex than MSL, an excessively large β focuses too much on learning disentangled representations, making it difficult to perform good data reconstruction.

4 CONCLUSION

We use the VAE-based disentanglement network for anomaly detection in multivariate time series. We find that high-quality disentangled representations can effectively improve anomaly detection performance. Our experiments on wide-studied time-series datasets show that our proposed method is simple and effective, outperforming all the strong baseline models while being more lightweight and interpretable.

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